
BIAS
DETECTION
IN NEWS MEDIA
FAIR AND
NEWS
TRUTH
NEWS
NEWS

PROJECT TOPIC:

BIAS DETECTION IN NEWS MEDIA

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PROBLEM INTRODUCTION

Problem Definition

- News outlets are often accused of biased reporting across dimensions - political, gender, religious, caste, regional, socioeconomic, and others.
- With increasing polarization and misinformation, unbiased reporting is critical.
- With the rapid growth of digital news, readers are exposed to vast and diverse information, often with subtle or overt biases.
- Biased reporting can influence public opinion and societal harmony.

Motivation

- There is a need for an automated, scalable solution that can analyze news articles for bias, offering transparent, multi-dimensional, and interpretable results.
- This project seeks to provide researchers, fact-checkers, and the public with actionable insights into media bias in Indian journalism.
- A bias detection system can help everyone identify and understand media bias, promoting transparency and informed decision-making.

ARCHITECTURE AND DESIGN

Design Goals

- Multi-Dimensional Detection: Support six bias dimensions.
- Scalability: Process large datasets both in batch and streaming modes.
- Robustness: Use an ensemble approach for accurate and interpretable results.
- Extensibility: Enable easy addition of new data sources, languages, and bias dimensions.
- Monitoring & Usability: Simplified deployment, monitoring, and result querying.

Key Features

- Automated web scraping from Indian news sources.
- Real-time and batch data processing pipelines.
- Storage and retrieval from a scalable database.
- Configurable ensemble bias detection.
- Monitoring UI, logs, error handling.

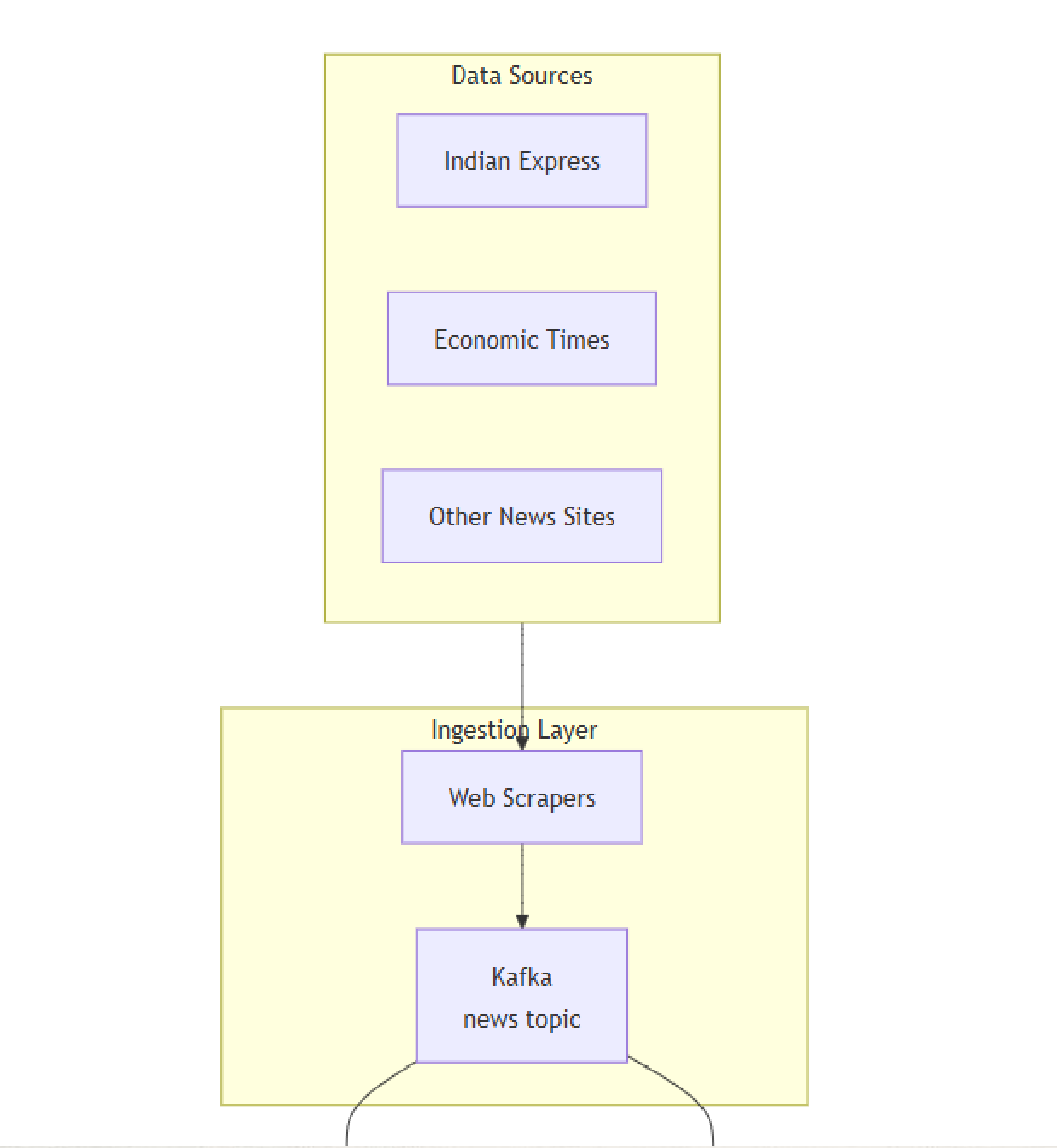
Data Collection

- Data was collected through the web scraping of the online news articles.
- To give an idea, we scraped articles from the "The Indian Express" from 1998 to 2025.
  - Took around 80 hours,
  - Using 4 Threads, where
  - Size per article was 4,000 to 8,000 words.
- Articles also collected from "The Economic Times" and "The Times of India".
  - Total number of samples that we eventually worked on was 303,518.

Data Flow

- Ingestion: Scrapers collect articles, push to Kafka.
- Processing: Spark jobs process data in batch (daily analysis) or stream (real-time analysis).
- ML Pipeline: Topic classification and bias detection using ensemble models.
- Storage: Results stored in MongoDB, logs/errors tracked for analysis and troubleshooting.

Data Flow Diagram



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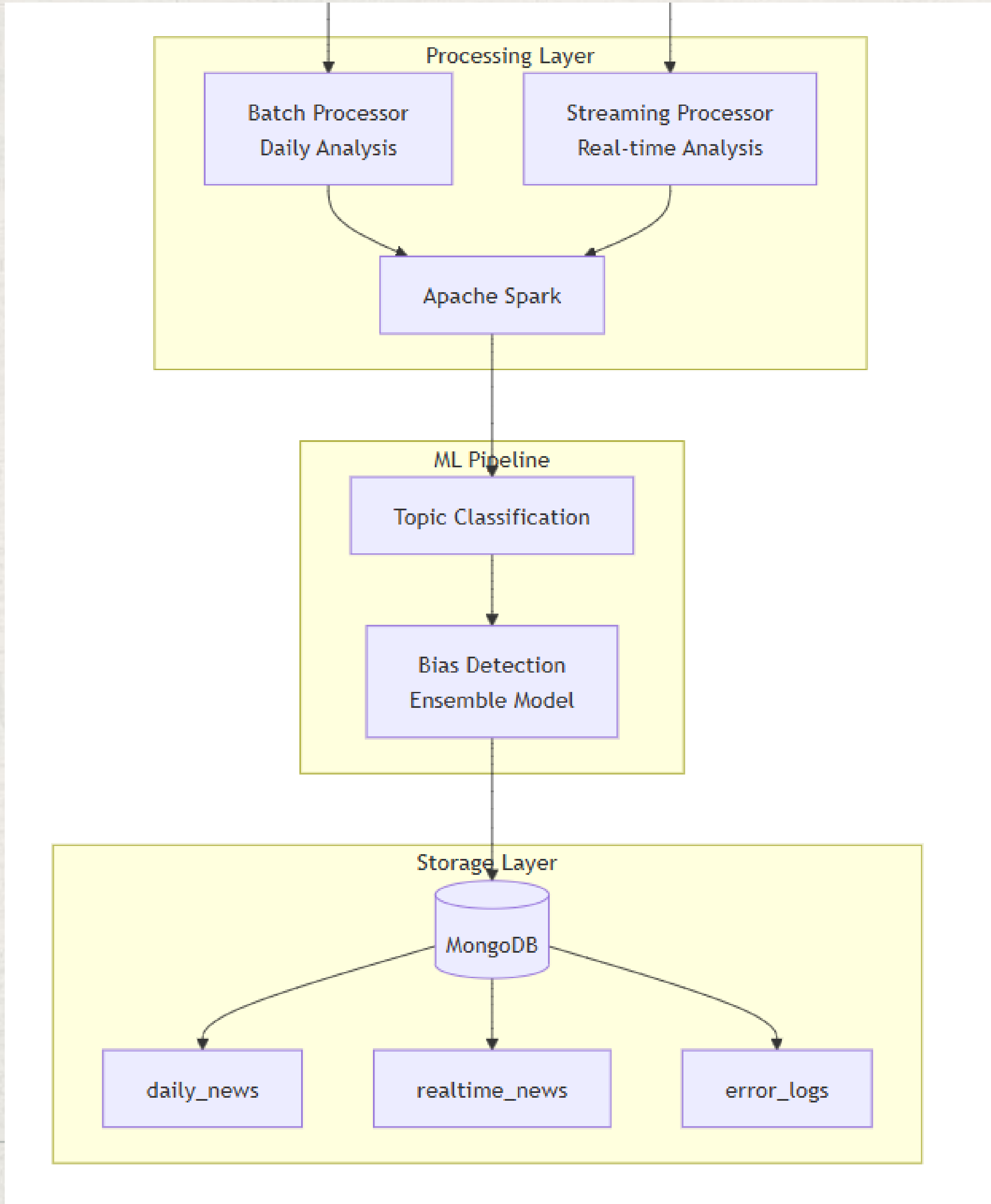
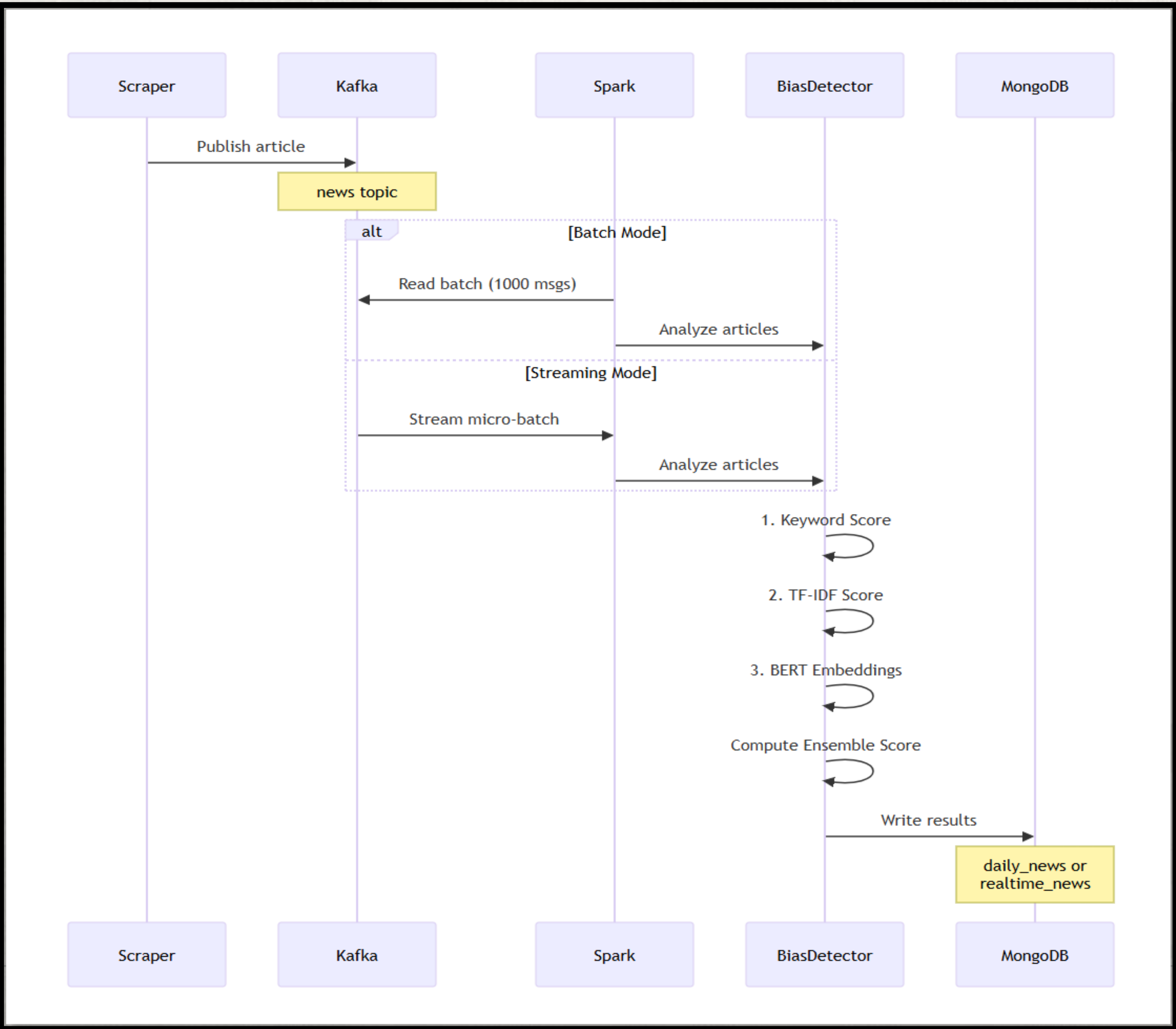
Data Flow Diagram



METHODOLOGY

Technologies

- Big Data Platforms Used:
  - Apache Spark for scalable distributed processing.
  - Apache Kafka for message queuing and streaming.
  - MongoDB for document-based storage and analytics.
- Orchestration: Docker, Docker-Compose for environment setup and deployment.
- Languages Used: Shell Scripting and Python.



Sequence Diagram

Machine Learning Methods

- Ensemble Approach (Bias Detection):
  - Keyword Matching (30% weight): Rule-based, interpretable
  - TF-IDF Analysis (30% weight): Statistical term importance
  - BERT Embeddings (40% weight): Semantic understanding (Sentence-Transformers)

Machine Learning Methods

- Topic Modeling & Classification:
    - Zero-shot classification
    - LDA topic modeling
    - TF-IDF clustering
    - Ensemble topic consensus

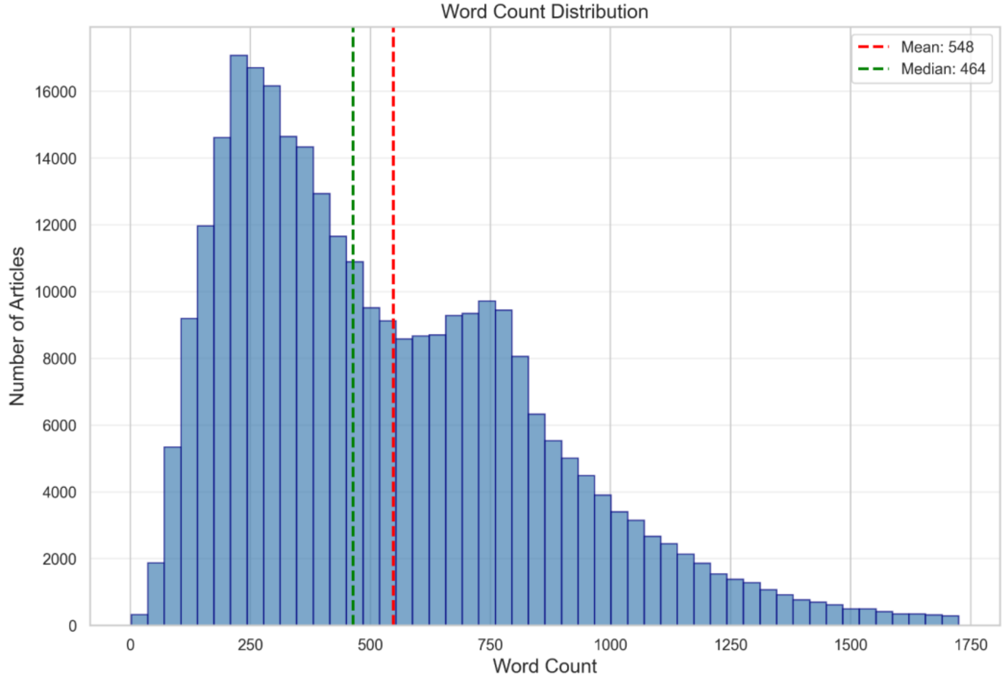
Scalability and Performance Metrics

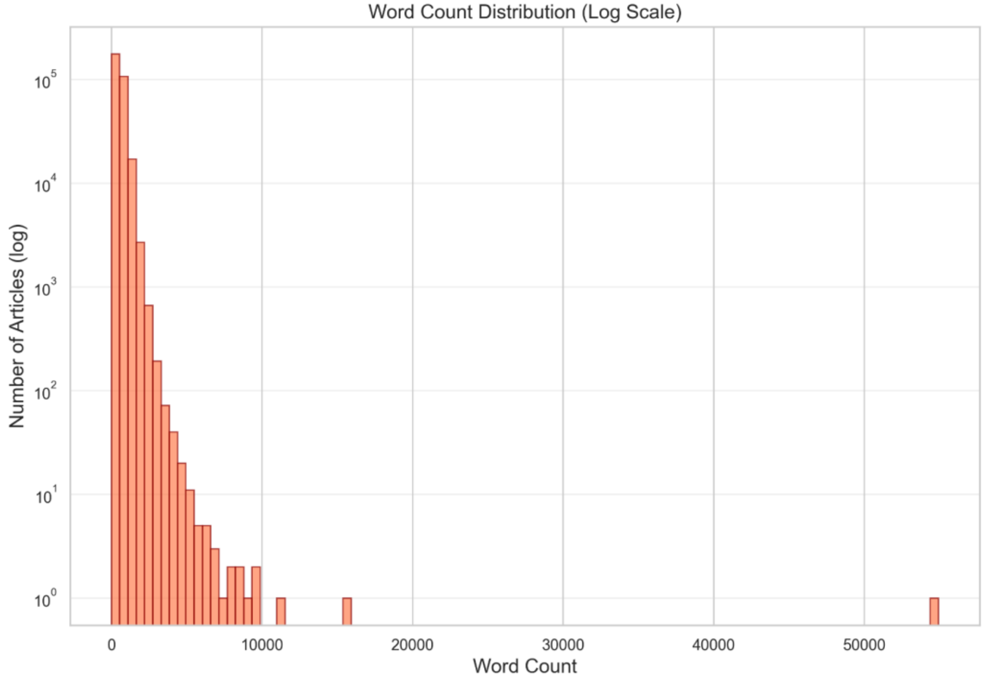
- End-to-End Latency: For the streaming pipeline, the time taken from when an article is published to Kafka until its analysis is stored in MongoDB.
- Throughput: The number of articles processed per second or minute.
- Resource Utilization: CPU and memory usage of the Spark cluster, which can be monitored via the Spark UI.

VISUALIZATIONS OF KEY RESULTS

Word Count Analysis Summary Dashboard

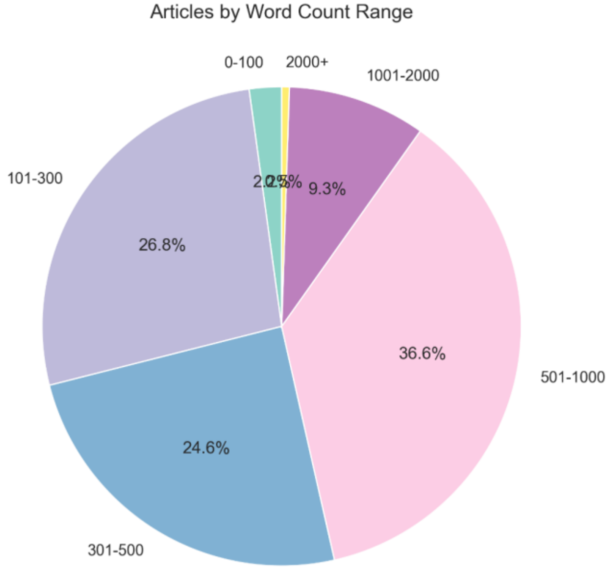
Word Count Analysis Summary



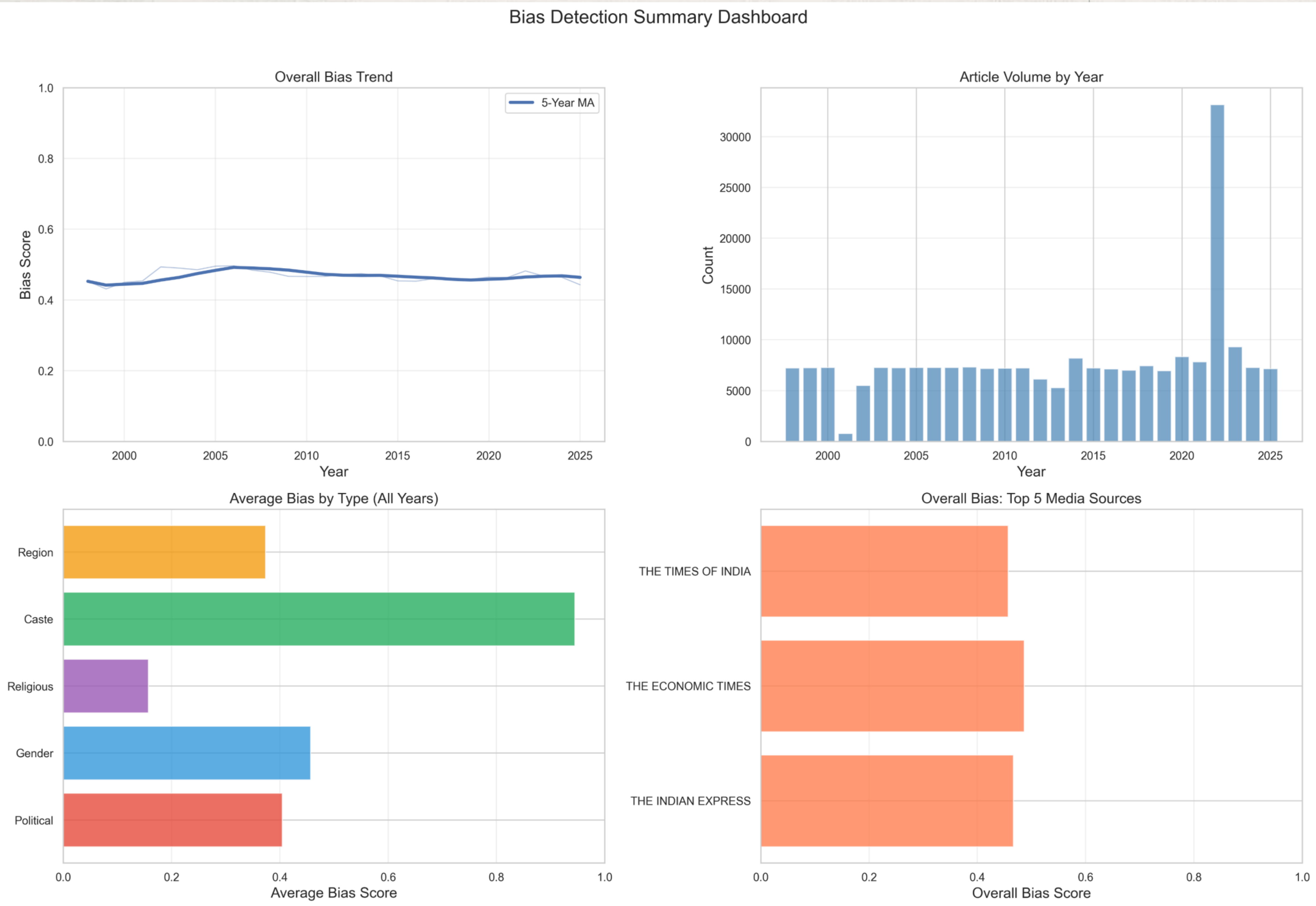


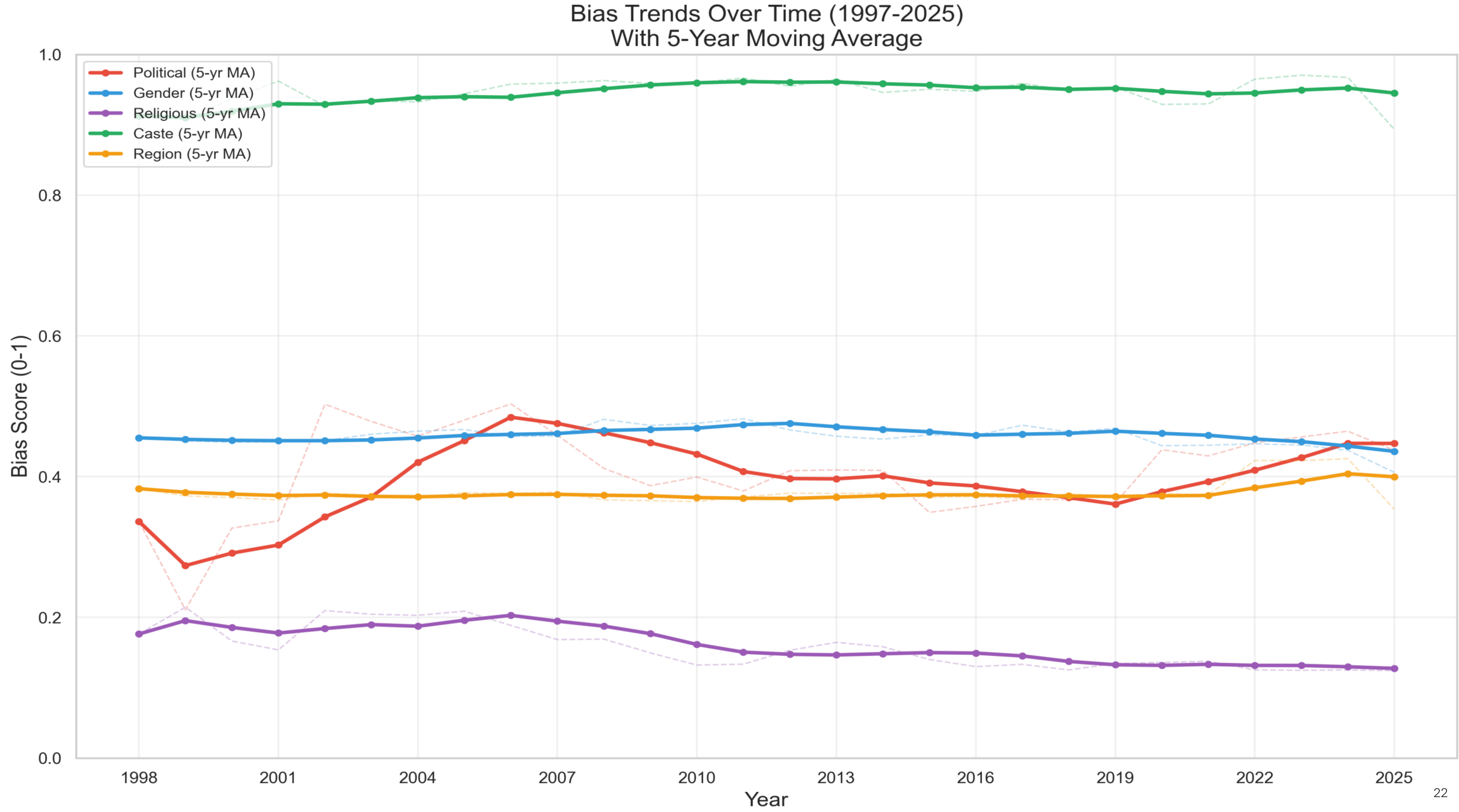
Word Count Statistics

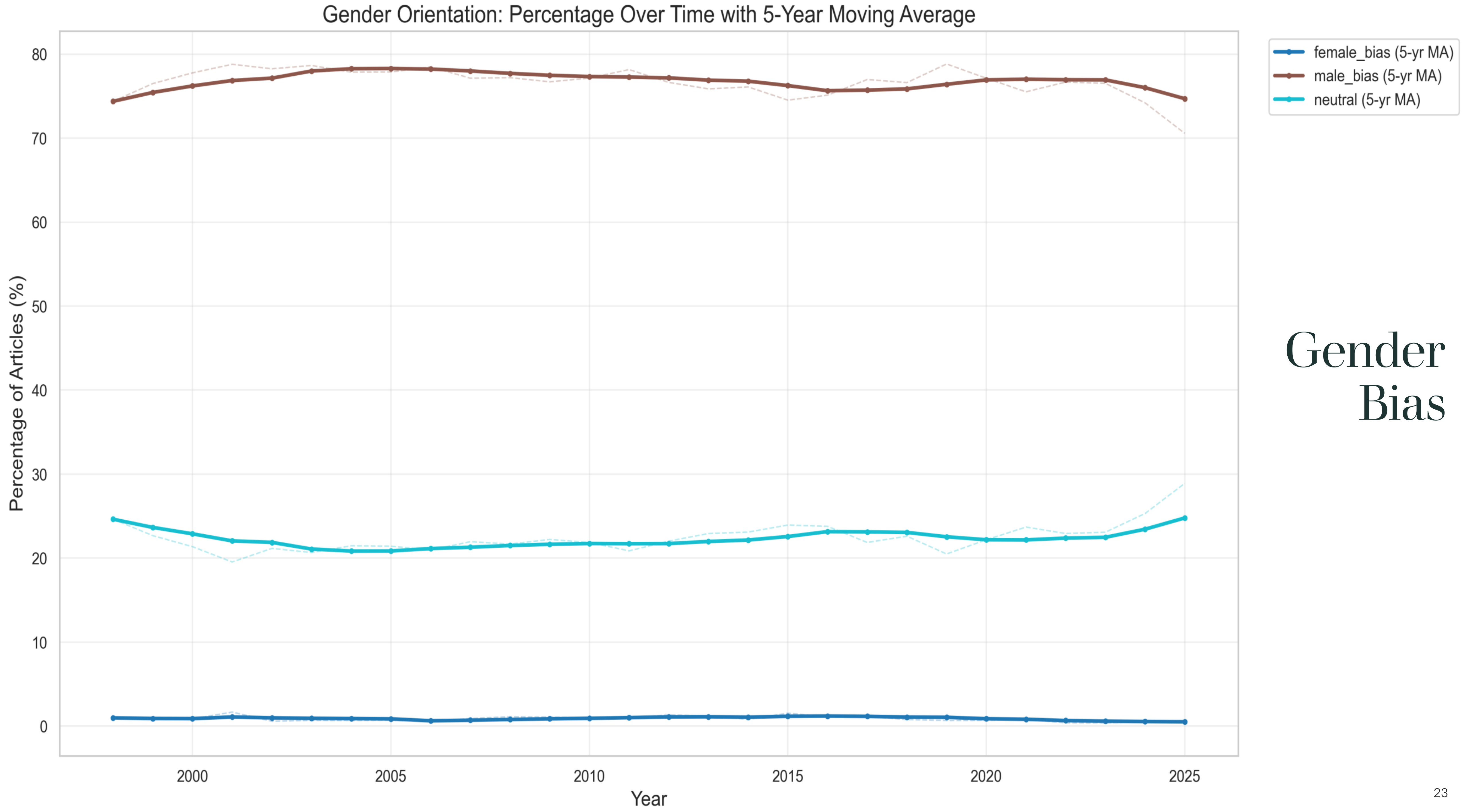
Metric	Value
Total Articles	303,101
Mean Word Count	548.1
Median Word Count	464.0
Std Dev	380.9
Min Word Count	2
Max Word Count	54,919
25th Percentile	275
75th Percentile	749
90th Percentile	996
99th Percentile	1725



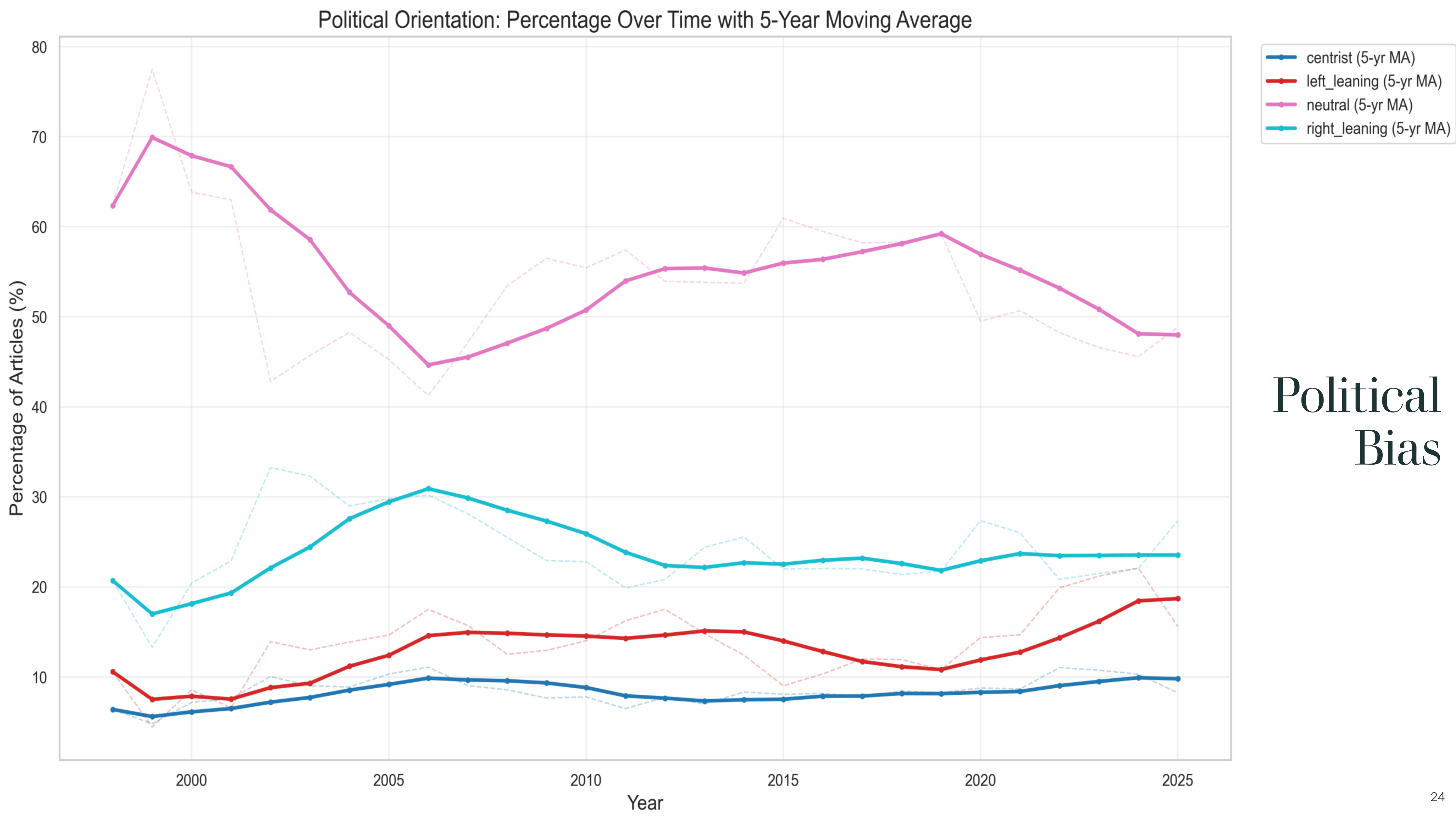
Bias Detection Summary Dashboard



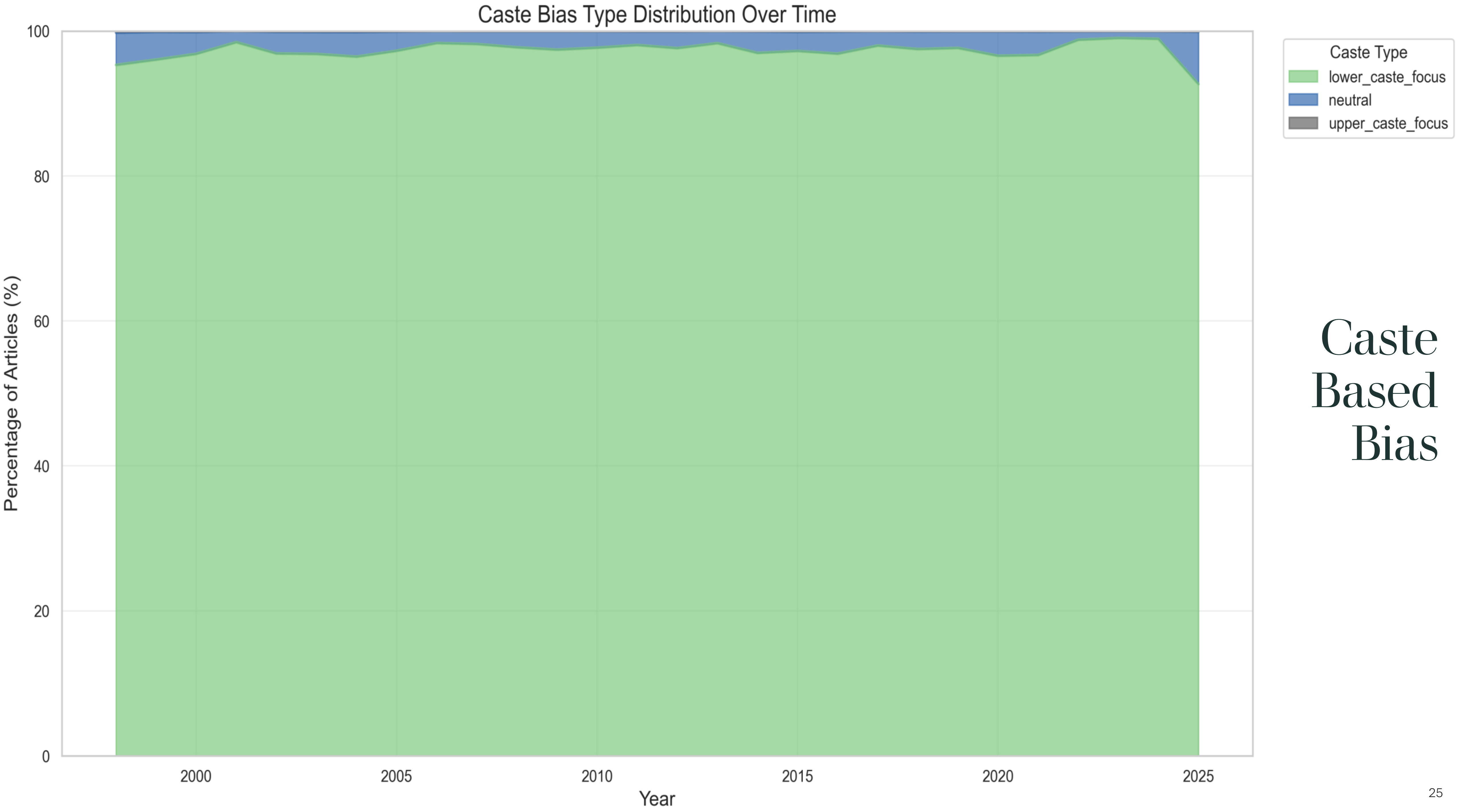

Bias Trends Over Time (1997-2025)
With 5-Year Moving Average
Political (5-yr MA)
Gender (5-yr MA)
Religious (5-yr MA)
Caste (5-yr MA)
Region (5-yr MA)
Bias Score (0-1)
Year
0.0
0.2
0.4
0.6
0.8
1.0
1998
2001
2004
2007
2010
2013
2016
2019
2022
2025


Gender Orientation: Percentage Over Time with 5-Year Moving Average
Percentage of Articles (%)
Year
0
10
20
30
40
50
60
70
80
2000
2005
2010
2015
2020
2025
female_bias (5-yr MA)
male_bias (5-yr MA)
neutral (5-yr MA)

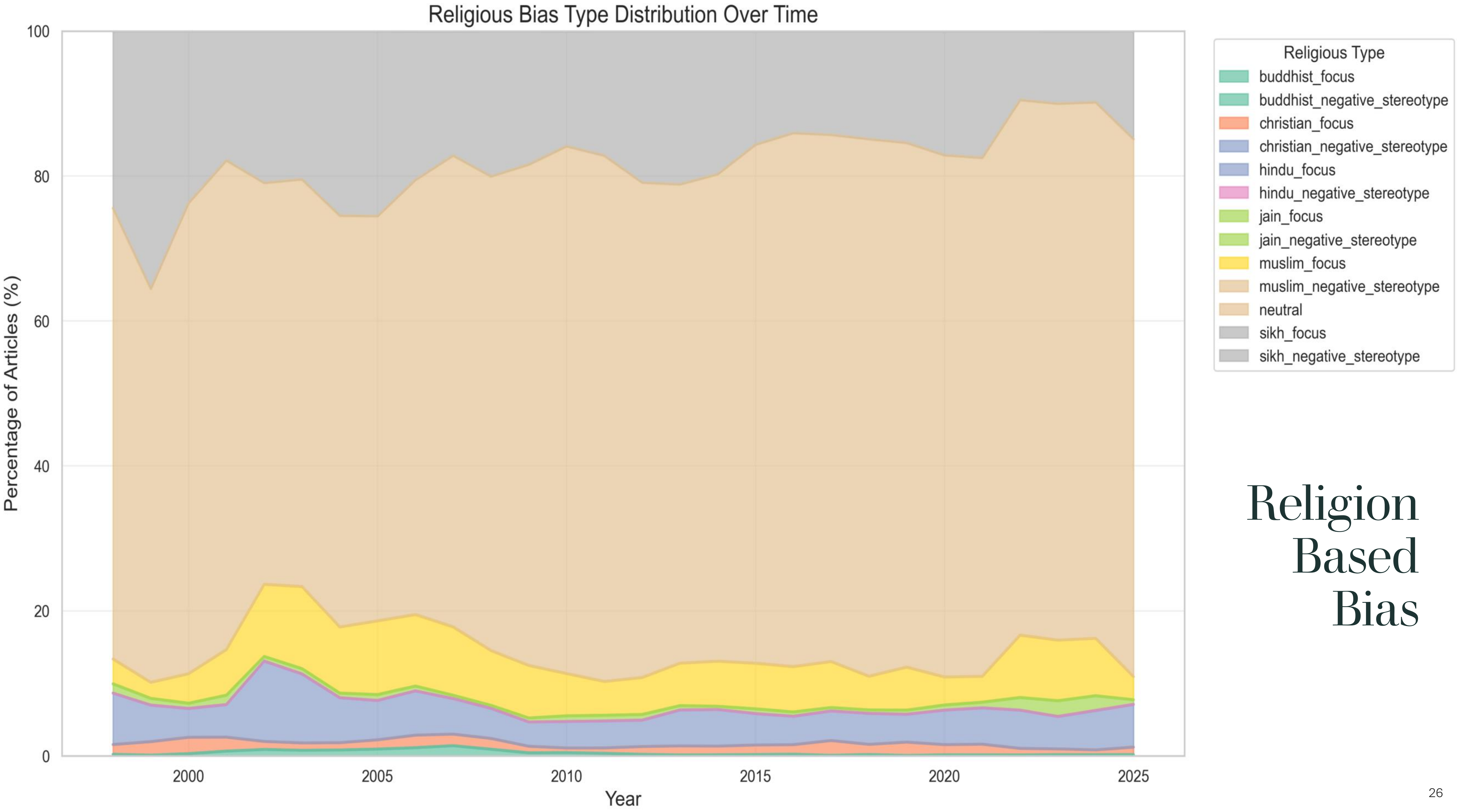
Gender Bias



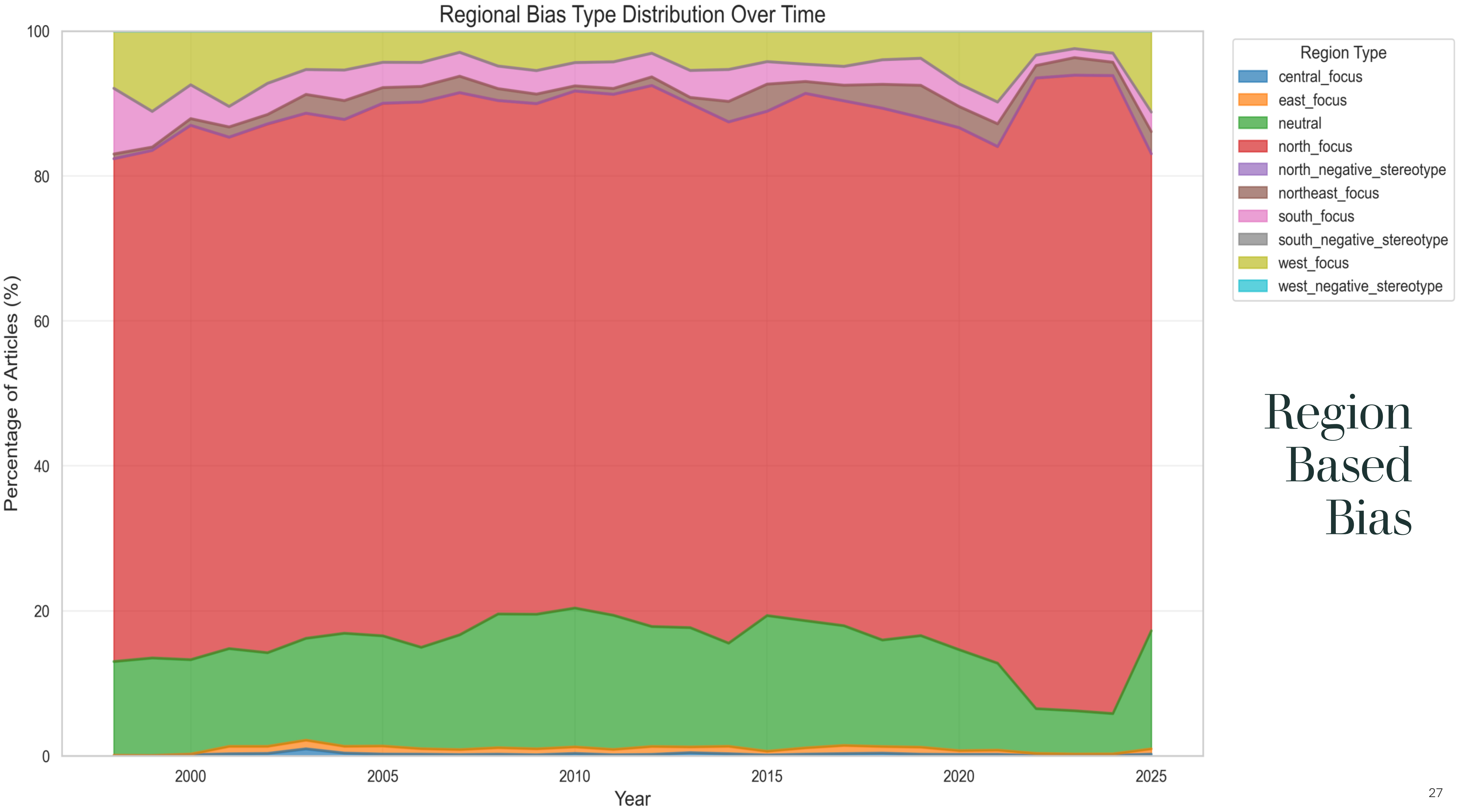
Political Bias



Caste Based Bias



Religion Based Bias



Region Based Bias

KEY PERFORMANCE BENCHMARKS

Batch Size = 1,000 | Topic Partition = 3

Data Set Size	Topic Partition	Producer Batch Size	Consumer Batch Size	Records Processed	Total Duration of Produce and Consume	Throughput	Average Batch Processing	Spark Cores and Memory
1 %	3	1000	1000	3035	58s	51 records/second	1238ms	2 Cores and 4 GB Memory
5 %	3	1000	1000	15175	178s	84 records/second	659ms	2 Cores and 4 GB Memory
10 %	3	1000	1000	30351	333s	91 records/second	663ms	2 Cores and 4 GB Memory
20 %	3	1000	1000	60703	629s	96 records/second	686ms	2 Cores and 4 GB Memory
50 %	3	1000	1000	151759	1539s	98 records/second	615ms	2 Cores and 4 GB Memory
100 %	3	1000	1000	303518	3065s	98 records/second	678ms	2 Cores and 4 GB Memory

Note: Total Data Size was 300,000 rows.

Batch Size = 10,000 | Topic Partition = 3

Data Set Size	Topic Partition	Producer Batch Size	Consumer Batch Size	Records Processed	Total Duration of Produce and Consume	Throughput	Average Batch Processing	Spark Cores and Memory
1 %	3	10000	10000	3035	44s	68 records/second	7187ms	2 Cores and 4 GB Memory
5 %	3	10000	10000	15175	48s	313 records/second	4195ms	2 Cores and 4 GB Memory
10 %	3	10000	10000	30351	59s	513 records/second	3788ms	2 Cores and 4 GB Memory
20 %	3	10000	10000	60703	88s	686 records/second	3554ms	2 Cores and 4 GB Memory
50 %	3	10000	10000	151759	183s	824 records/second	4563ms	2 Cores and 4 GB Memory
100 %	3	10000	10000	303518	339s	895 records/second	5152ms	2 Cores and 4 GB Memory

Note: Total Data Size was 300,000 rows.

Batch Size = 10,000 | Topic Partition = 6

Data Set Size	Topic Partition	Producer Batch Size	Consumer Batch Size	Records Processed	Total Duration of Produce and Consume	Throughput	Average Batch Processing	Spark Cores and Memory
1 %	6	10000	10000	3035	43s	69 records/second	3996ms	2 Cores and 4 GB Memory
5 %	6	10000	10000	15175	48s	310 records/second	4670ms	2 Cores and 4 GB Memory
10 %	6	10000	10000	30351	63s	477 records/second	3295ms	2 Cores and 4 GB Memory
20 %	6	10000	10000	60703	93s	650 records/second	3141ms	2 Cores and 4 GB Memory
50 %	6	10000	10000	151759	188s	804 records/second	2767ms	2 Cores and 4 GB Memory
100 %	6	10000	10000	303518	338s	896 records/second	3019ms	2 Cores and 4 GB Memory

Note: Total Data Size was 300,000 rows.

Batch Size = 10,000 | Topic Partition = 6

Varying the Spark Cores and Memory

Data Set Size	Topic Partition	Producer Batch Size	Consumer Batch Size	Records Processed	Total Duration of Produce and Consume	Throughput	Average Batch Processing	Spark Cores and Memory
100 %	6	10000	10000	303518	338s	896 records/second	3019ms	2 Cores and 4 GB Memory
100 %	6	10000	10000	303518	334s	908 records/second	3682ms	4 Cores and 4 GB Memory
100 %	6	10000	10000	303518	328s	923 records/second	3051ms	6 Cores and 4 GB Memory

Note: Total Data Size was 300,000 rows.

SUMMARY

What We Achieved

- The system meets its design goals:
  - multi-dimensional bias detection,
  - scalable batch/streaming pipelines,
  - robust ensemble methodologies, and
  - modular/extensible architecture.
- Provides actionable insights and analytics on Indian news media bias, aligning with original motivation and proposal.

Envisioned Scale of Data

- A conservative, evidence-based estimate:
  - Large/global publishers (top ~1,000): average ~50–200 articles/day → 50,000–200,000/day.
  - Mid/small publishers and local outlets (tens of thousands): if 20,000 sites average 5–20/day → 100,000–400,000/day.
  - Automated and niche feeds, blogs, and micro-sites (many thousands): likely add another 100,000–500,000/day.

Envisioned Scale of Data

- More aggressive estimate (including broad blogosphere, social-news posts, and automated micro-updates):
  - 1 million to 3+ million per day is plausible depending on inclusion of short automated items and social-origin reporting.

Roadmap and Future Work

- Source Expansion: More Indian news outlets and support for regional languages.
- Cloud Deployment: Migrate to AWS (EMR, MSK, DynamoDB).
- Real-time Alerting: Automated notifications for high-bias content.
- Dashboard Visualization: Web-based dashboards for broader user outreach.
- Community Feedback and Further Research: Incorporate impact measurements and retrain models.

THANK YOU!